

Given: Due:	Further Maths (Decision) HW 05				Name:	
Chapter 2: Graph theory						
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.						
91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)	
Useful resources: Useful resources: Your notes from recent lessons & your text book						
Questions - do these on lined paper						
(a) Explain why it is not possible to draw a graph with exactly 5 nodes with orders 1, 3, 4, 4 and 5 (1) A connected graph has exactly 5 nodes and contains 18 arcs. The orders of the 5 nodes are $2^{2x} - 1$, 2^x, $x + 1$, $2^{x+1} - 3$ and $11 - x$ (b) (i) Calculate x. (ii) State whether the graph is Eulerian, semi-Eulerian or neither. You must justify your answer. (6) (c) Draw a graph which satisfies all of the following conditions: - The graph has exactly 5 nodes. - The nodes have orders 2, 2, 4, 4 and 4 - The graph is not Eulerian. (2)						
(a) Draw the graph K_5 (1) (b) (i) In the context of graph theory explain what is meant by 'semi-Eulerian'. (ii) Draw two semi-Eulerian subgraphs of K_5, each having five vertices but with a different number of edges. (3) (c) Explain why a graph with exactly five vertices with vertex orders 1, 2, 2, 3 and 4 cannot be a tree. (2)						
END OF HOMEWORK						
Total = 15 marks						
Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.						
File this question paper in your file, ready for revision and file inspection.						